

Darukaa.Earth Knowledge Synthesis Report 08

Land-Use Change, Grasslands, Forests and Agricultural Conversion

Document type: institutional scientific synthesis

Topics: land_cover, grassland, forest, agricultural land, land-use change

1. Purpose

This report describes how conversion among forest, grassland, agricultural land, and urban land changes habitat diversity and soil function. It exists so the knowledge base can answer land-cover questions without collapsing all land use into 'farming'.

2. Grasslands

Natural and semi-natural grasslands can hold high plant diversity and soil carbon, especially in roots. Conversion to monoculture cropland typically reduces plant diversity and can deplete SOC if tillage and residue removal are intense. Overgrazing also degrades grasslands via cover loss and compaction.

3. Forests

Forests store carbon in biomass and soils and provide structurally complex habitat. Conversion to agriculture or urban land is habitat destruction. Selective logging and degradation also fragment function even if a forest class remains on a map.

4. Agricultural land

Agriculture can support biodiversity if it retains mixed cover, wetlands, trees, and low chemical pressure. It becomes a biodiversity sink when it is simplified, dry, nutrient-depleted, and chemically stressed at once. Multi-metric assessment is therefore required: land-cover class plus soil plus climate plus pollution.

5. Urban land

See the companion pollution and urban report. Sealing is effectively irreversible on human timescales for soil biodiversity unless surfaces are removed.

6. Source basis

FAO land-cover and soil carbon discussions; IPBES land-use change synthesis; UNCCD degradation hierarchy of avoid?reduce?restore. Pages refer to this Darukaa document.